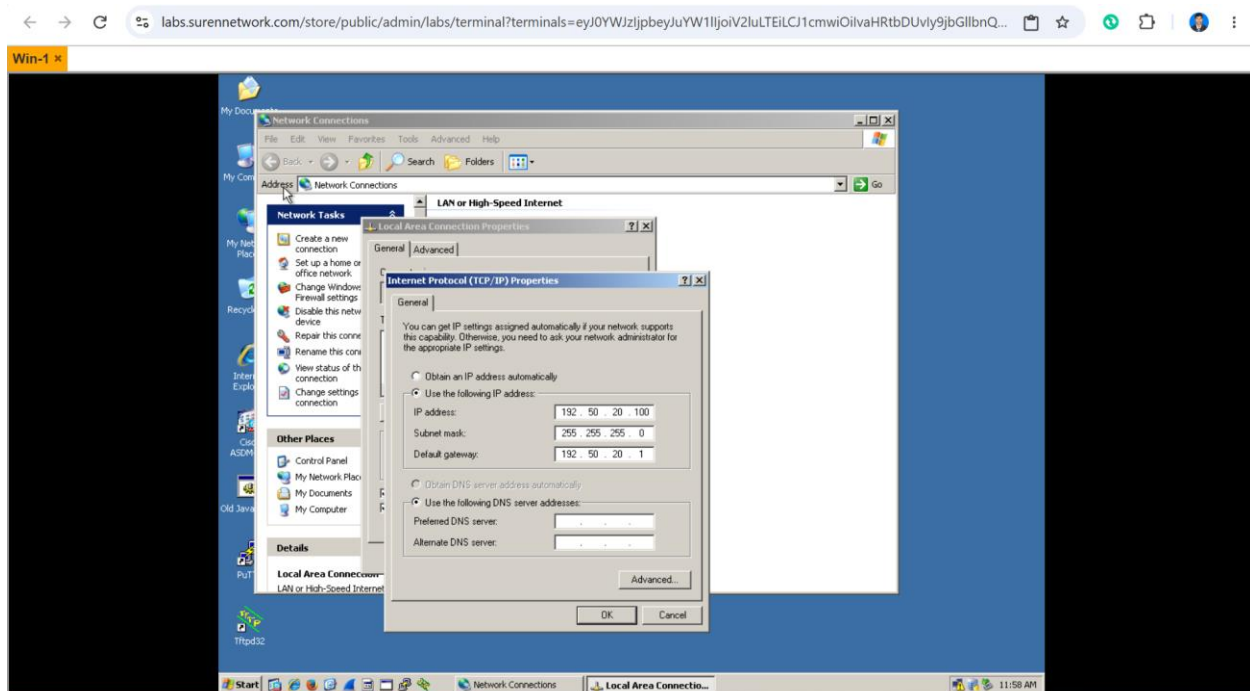




```

Win-1 x R1 x
R1#
R1#config term
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#interface e0/0
R1(config-if)#
R1(config-if)#ip address 192.50.20.1 255.255.255.0
R1(config-if)#
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#exit
R1(config)#
R1(config)#interface e0/1
R1(config-if)#
R1(config-if)#ip address 12.0.0.1 255.0.0.0
R1(config-if)#
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#exit
R1(config)#
R1(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              192.50.20.1    YES manual up          up
Ethernet0/1              12.0.0.1       YES manual up          up
Ethernet0/2              unassigned     YES TFTP  down        down
Ethernet0/3              unassigned     YES TFTP  down        down
R1(config)#
*Aug 22 11:00:47.723: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#

```

```

Win-1 x R1 x
Ethernet0/2              unassigned     YES TFTP  down        down
Ethernet0/3              unassigned     YES TFTP  down        down
R1(config)#
*Aug 22 11:00:47.723: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 12.0.0.2
R1(config)#
R1(config)#exit
R1#
R1#
*Aug 22 11:08:05.027: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 12.0.0.2 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 12.0.0.2
     12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     12.0.0.0/8 is directly connected, Ethernet0/1
L     12.0.0.1/32 is directly connected, Ethernet0/1
     192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.50.20.0/24 is directly connected, Ethernet0/0
L     192.50.20.1/32 is directly connected, Ethernet0/0
R1#

```

```
Win-1 x R1 x R2 x
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
R2#
*Aug 22 11:12:22.536: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    12.0.0.2        YES manual up          up
Ethernet0/1    26.0.0.1        YES manual up          up
Ethernet0/2    23.0.0.1        YES manual up          up
Ethernet0/3    24.0.0.1        YES manual up          up
R2#
R2#config term
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#ip route 192.50.20.0 255.255.255.0 12.0.0.1
R2(config)#
R2(config)#ip route 223.0.50.0 255.255.255.0
*Aug 22 11:14:23.772: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.2
R2(config)#
R2(config)#ip route 32.0.0.0 255.0.0.0 23.0.0.2
R2(config)#
R2(config)#ip route 45.0.0.0 255.0.0.0 24.0.0.2
R2(config)#
R2(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
R2(config)#
R2(config)#exit
R2#s
*Aug 22 11:16:04.464: %SYS-5-CONFIG_I: Configured from console by console
```

```
Win-1 x R1 x R2 x
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
 23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/2
L    23.0.0.1/32 is directly connected, Ethernet0/2
 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/3
L    24.0.0.1/32 is directly connected, Ethernet0/3
 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.1/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
S    172.36.0.0/16 [1/0] via 23.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2
R2#
R2#
```

```

Win-1 x R1 x R2 x R3 x
R3#
R3#config term
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#interface e0/0
R3(config-if)#
R3(config-if)#ip address 23.0.0.2 255.0.0.0
R3(config-if)#
R3(config-if)#no shutdown
R3(config-if)#
R3(config-if)#exit
R3(config)#
R3(config)#interface e0/1
R3(config-if)#
R3(config-if)#ip address 32.0.0.1 255.0.0.0
R3(config-if)#
R3(config-if)#no shutdown
R3(config-if)#
R3(config-if)#exit
R3(config)#
R3(config)#exit
R3#
*Aug 22 11:19:37.335: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	23.0.0.2	YES	manual	up	up
Ethernet0/1	32.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R3#

```

```

*Aug 22 11:19:37.335: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 23.0.0.1
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 23.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 [1/0] via 32.0.0.2
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1
R3#
R3#
R3#

```

```
R4#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    24.0.0.2        YES manual up          up
Ethernet0/1    45.0.0.1        YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down
R4#
```

```
Win-1 x R1 x R2 x R3 x R4 x
R4#
R4#exit
*Aug 22 11:35:16.166: %SYS-5-CONFIG_I: Configured from console by console
R4#
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 24.0.0.1
     24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/0
L    24.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 45.0.0.2
     45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/1
L    45.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
S    223.0.50.0/24 [1/0] via 24.0.0.1
R4#
R4#
```

```

R5#
R5#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    172.36.0.1      YES manual up          up
Ethernet0/1    32.0.0.2        YES manual up          up
Ethernet0/2    45.0.0.2        YES manual up          up
Ethernet0/3    unassigned      YES TFTP  down        down
R5#

```

```

Win-1 x R1 x R2 x R3 x R4 x R5 x
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.2/32 is directly connected, Ethernet0/1
    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/2
L    45.0.0.2/32 is directly connected, Ethernet0/2
    172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.36.0.0/16 is directly connected, Ethernet0/0
L    172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
R5#
R5#
R5#

```

Win-1 x R1 x R2 x R3 x R4 x R5 x R6 x

```
R6#
R6#config term
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#
R6(config)#interface e0/0
R6(config-if)#
R6(config-if)#ip address 223.0.50.1 255.255.255.0
R6(config-if)#
R6(config-if)#no shutdown
R6(config-if)#
R6(config-if)#exit
R6(config)#
R6(config)#interface e0/1
R6(config-if)#
R6(config-if)#ip address 26.0.0.2 255.0.0.0
R6(config-if)#
R6(config-if)#no shutdown
R6(config-if)#
R6(config-if)#exit
R6(config)#
R6(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              223.0.50.1     YES manual up            up
Ethernet0/1              26.0.0.2       YES manual up            up
Ethernet0/2              unassigned     YES TFTP  down          down
Ethernet0/3              unassigned     YES TFTP  down          down
R6(config)#
```

Win-1 x R1 x R2 x R3 x R4 x R5 x R6 x

```
*Aug 22 11:54:09.315: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config)#
R6(config)#
R6(config)#ip route 0.0.0.0 0.0.0.0 26.0.0.1
R6(config)#
R6(config)#exit
R6#
*Aug 22 11:55:16.596: %SYS-5-CONFIG_I: Configured from console by console
R6#
R6#
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 26.0.0.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 26.0.0.1
      26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      26.0.0.0/8 is directly connected, Ethernet0/1
L      26.0.0.2/32 is directly connected, Ethernet0/1
      223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C      223.0.50.0/24 is directly connected, Ethernet0/0
L      223.0.50.1/32 is directly connected, Ethernet0/0
R6#
R6#
```

